ATP, Photosynthesis, & Cellular Respiration

STEM Success CRAM 11/2

ATP

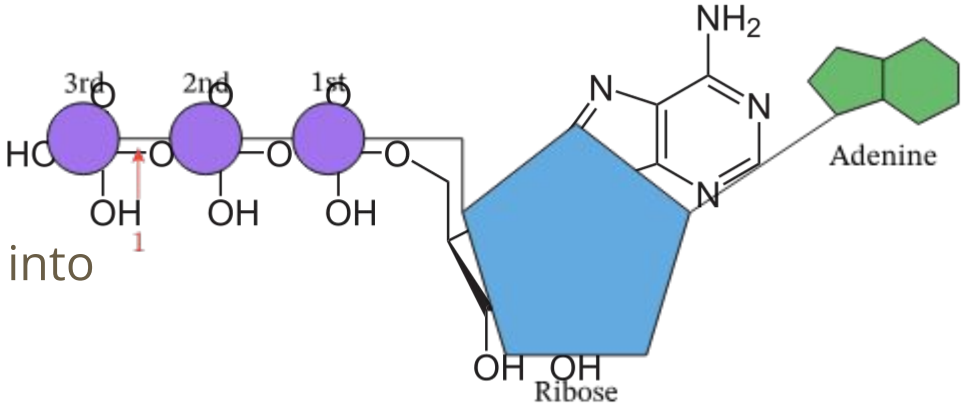
Adenosine Triphosphate (ATP) → the energy currency of the cell

Structure:

- Adenine
- Ribose
- Phosphate Group

“Fuel” from Glucose is converted into

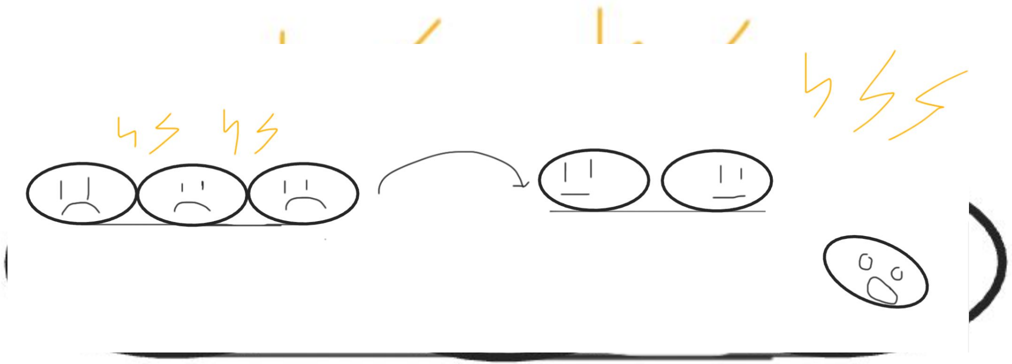
ATP to be used by the cell

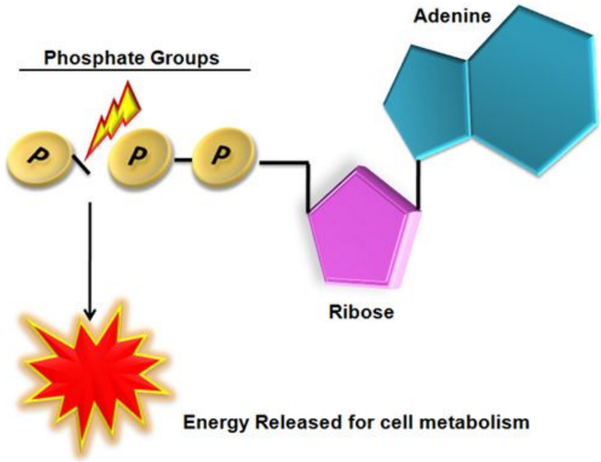


The Energy Currency of the Cell

Phosphate bonds are very unstable, and energy is released by breaking the last phosphate from our ATP molecule

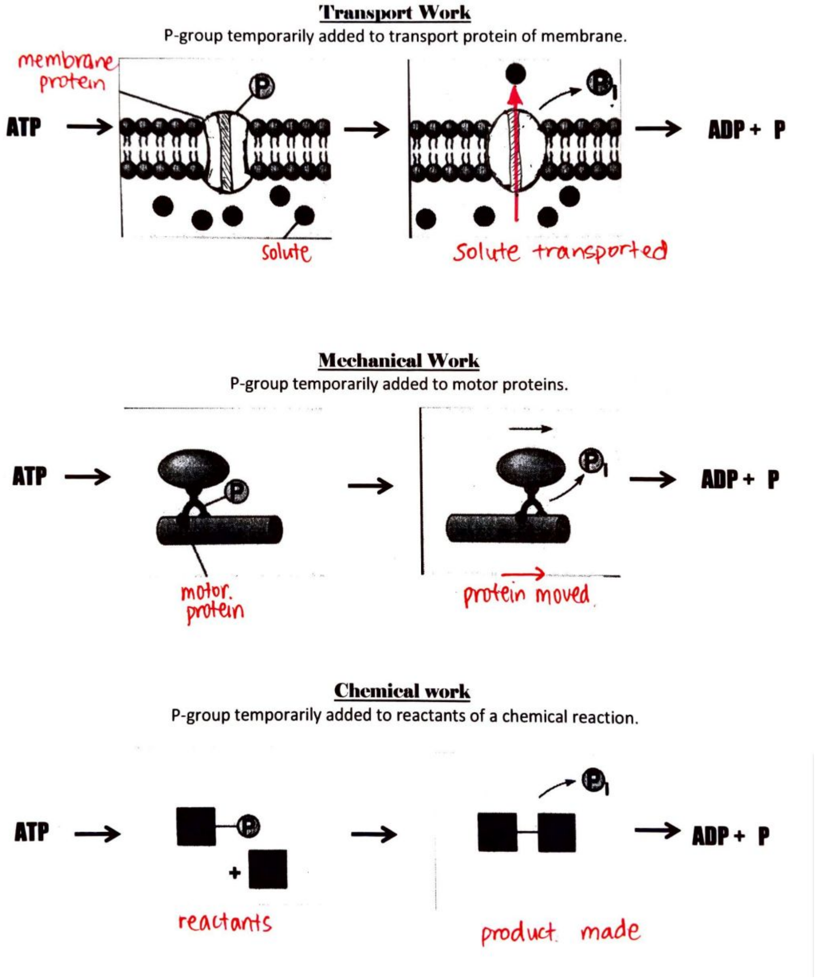
- Phosphate Groups are negatively charged and repel each other in ATP
    - Held Together by Phosphoanhydride Bonds
- Three Guys on a Bench





Uses of ATP

- Chemical Work
  - Chemical Reactions
  - Cell Respiration
  - Photosynthesis
- Mechanical Work
  - Muscle Contraction
  - Movement of organelles in cells
- Transport Work
  - Active transport



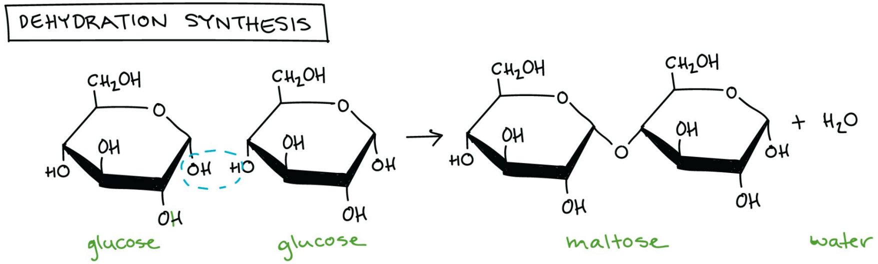
Hydrolysis and Dehydration Synthesis

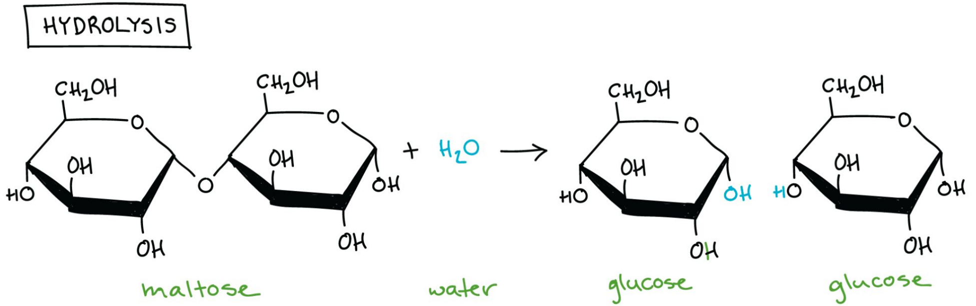
Hydrolysis → chemical breakdown of a compound due to a reaction with water

- $ATP + H_2O \rightarrow ADP + P_i$ + energy; proteins breaking down into amino acids

Dehydration Synthesis → creation of a larger molecule from smaller monomers where a water molecule is released

- $ADP + P_i$ + energy $\rightarrow H_2O + ATP$; glucose molecules coming together to form starch





Other Types of Energy Storages

ADP → Low energy: Adenosine Diphosphate

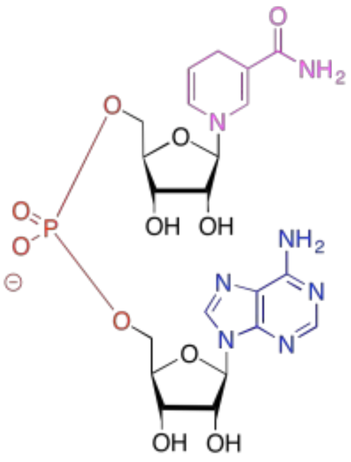
- AMP → Adenosine Monophosphate

NADPH → Used in Photosynthesis

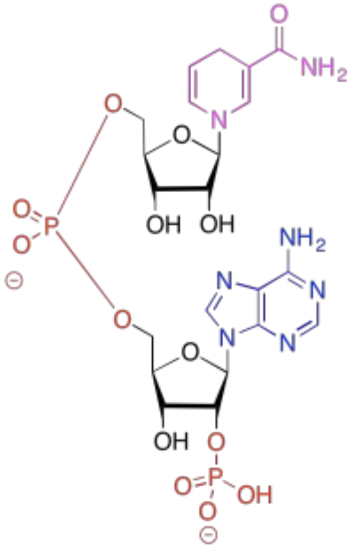
NADH

$FADH_2$

Used in Cellular Respiration



NADH



NADPH

Photosynthesis

Photosynthesis is the process by which plants convert solar energy (light) into glucose that can be used by the plants or other organisms on Earth.

The chemical equation of photosynthesis is the following:

$6CO_2 + 6H_2O \rightarrow C_6H_{12}O_6 + 6O_2$

which is the opposite of the chemical equation for cellular respiration.

Pigments

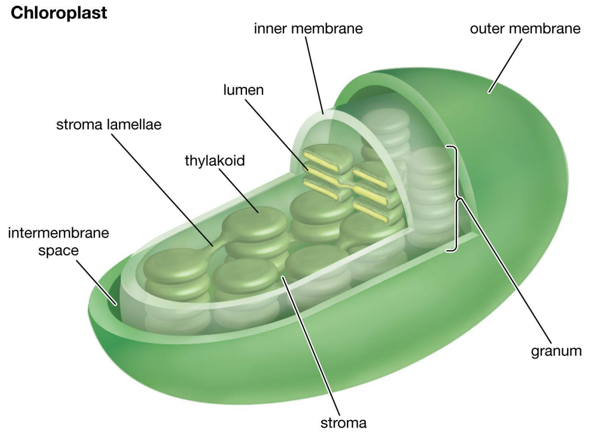
Pigments- molecules that can capture light

Plants use pigments such as chlorophyll to absorb light.

Chlorophyll absorbs red and blue light rays but reflects green wavelengths hence why many plants appear green to our eyes. Other pigments exist in plants as well, hence why plants don't always appear green. Chlorophyll are located in the chloroplast of a plant cell.

Chloroplasts are important because the the two processes of photosynthesis, light dependent and light independent reactions depend on the chloroplast.

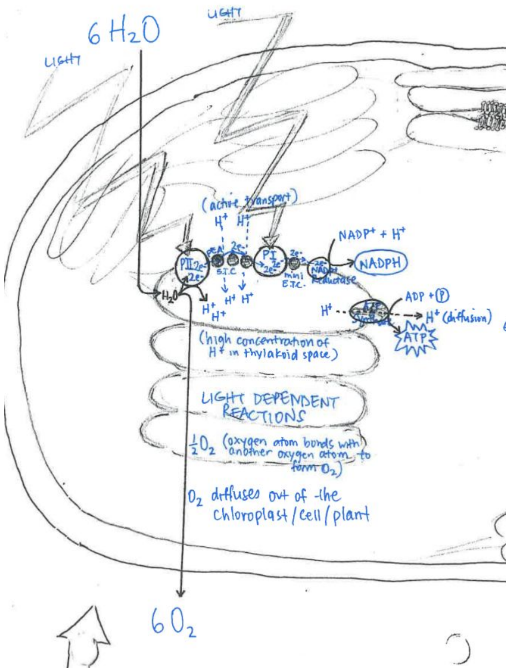
Important: The color an object appears is determined by the colors of the lightwaves it reflects.



Light Dependent Reactions

Light Dependent Reactions occur in the thylakoid of the chloroplast and use light energy and water to produce the two molecules needed for the Light Independent reactions: ATP (energy storage) and NADPH (electron carrier).

Photosystems are protein and pigment structures designed to harvest light.



Light Dependent Reactions cont.

When light is captured by the pigments in photosystem 2, light energy is transferred from pigment to pigment until it reaches photosystem 1.

While the energy is being transferred, hydrogen ions are actively transported into the chloroplast and water is also split into hydrogen ions and oxygen ions inside the chloroplast. The active transport of hydrogen ions is driven by the loss of energy of the electrons in the photosystems as they travels through the transport chain with the light.

A singular oxygen ion from a water molecule will eventually bond with another before diffusing out of the cell.

Light Dependent Reactions cont.

The hydrogen ions that were actively transported into the cell eventually diffuse out of the cell through the ATP synthase forming ATP.

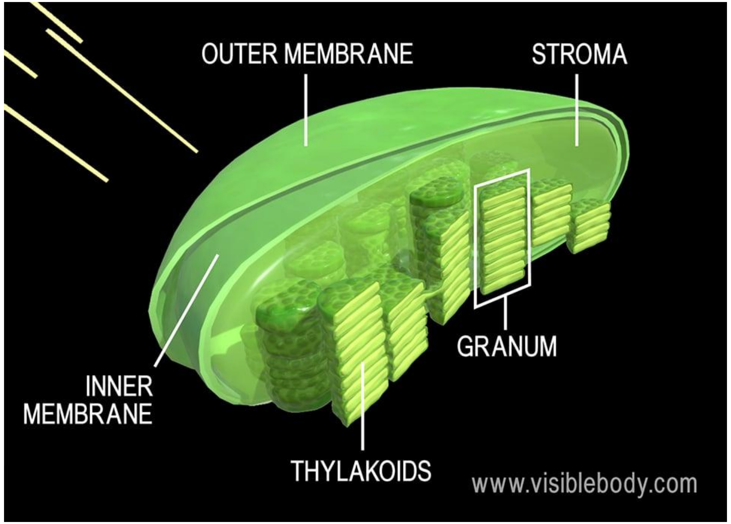
When energy reaches photosystem 1, the electron in photosystem 1 gets boosted to an electron acceptor, and the low-energy electron from photosystem 2 ends up replacing the boosted electron.

The high energy electron from photosystem 1 goes down another electron transport chain where it reaches NADP+ and forms NADPH.

Light Independent Reactions

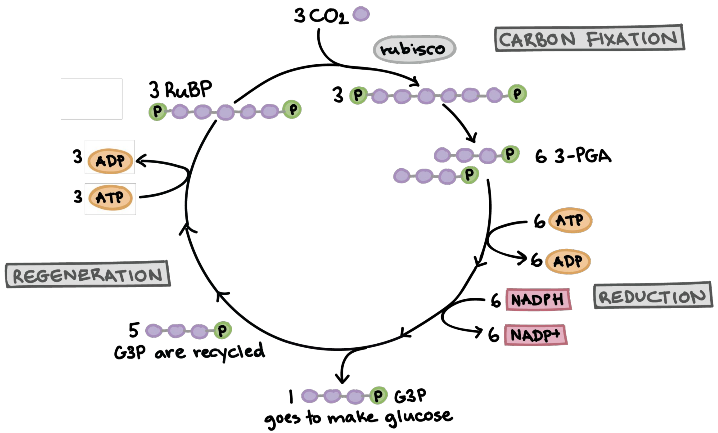
The Light Independent Reactions occur in the stroma- the inner space of the chloroplasts- and is the process that actually makes glucose out of carbon and the products of the light dependent reactions.

The Light Independent Reactions are also known as the Calvin cycle and have 3 main stages.



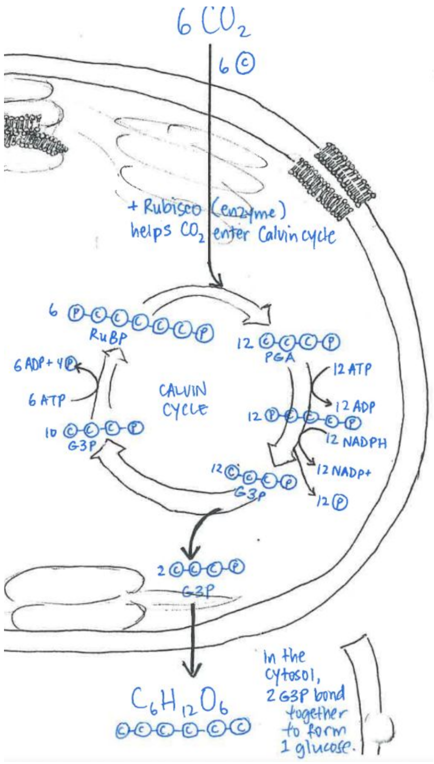
Light Independent Reactions- Carbon Fixation

A CO2 molecule enters a leaf through the stomata and diffuses into the stroma. With the help of the enzyme rubisco, it combines with an acceptor molecule that has 5 carbons (RuBP). The acceptor molecule now has 6 carbons, which eventually splits into 12 3 carbon strings (3-PGA or phosphoglyceric acid).



Light Independent Reactions- Reduction

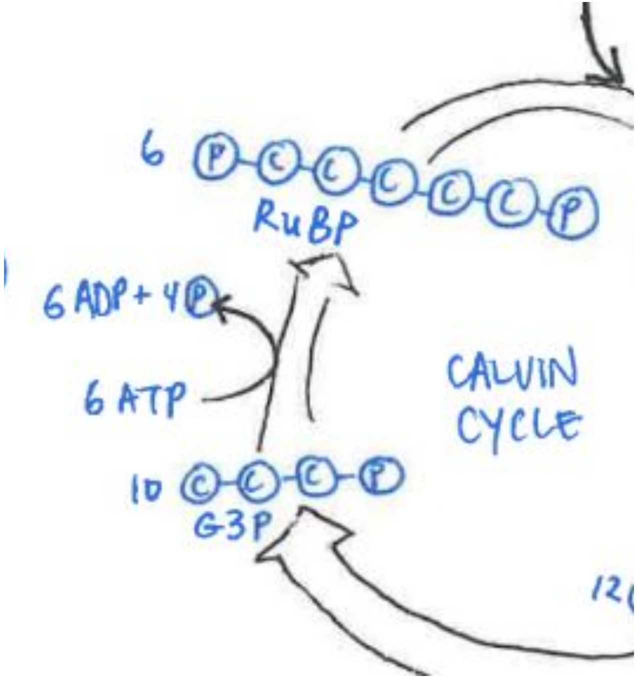
Now, the products of the light dependent reactions- NADPH and ATP- convert 3-PGA into glyceraldehyde-3-phosphate (G3P). However, in this process NADPH loses electrons and ATP loses a phosphate group, hence the name. 12 G3P molecules are formed but only 2 help create glucose.



Light Independent Reactions- Regeneration

While some G3P molecules make glucose the rest must be recycled so that the RuBP acceptor can be regenerated. The remaining 10 G3P molecules are eventually reformed into 2 RuBP by ATP hydrolysis and the addition of 4 phosphate groups.

Important: these processes are constantly occurring!



Cellular Respiration

Cellular respiration is a process that converts glucose into ATP.

The chemical formula for this process is:

$C6H12O6 + 6O2 = 6CO2 + 6H2O + ATP$

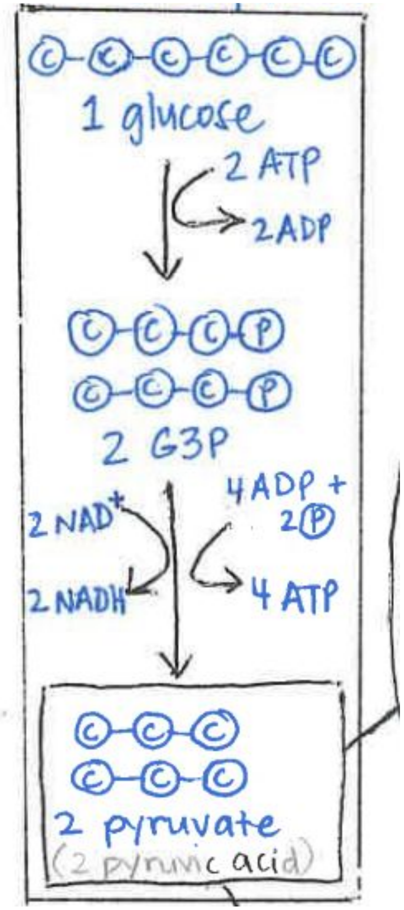
(glucose) + (oxygen) = (carbon dioxide) + (water) + ATP

**reactants and products are opposite to those of photosynthesis!

3 main parts include: glycolysis, Krebs cycle, electron transport chain

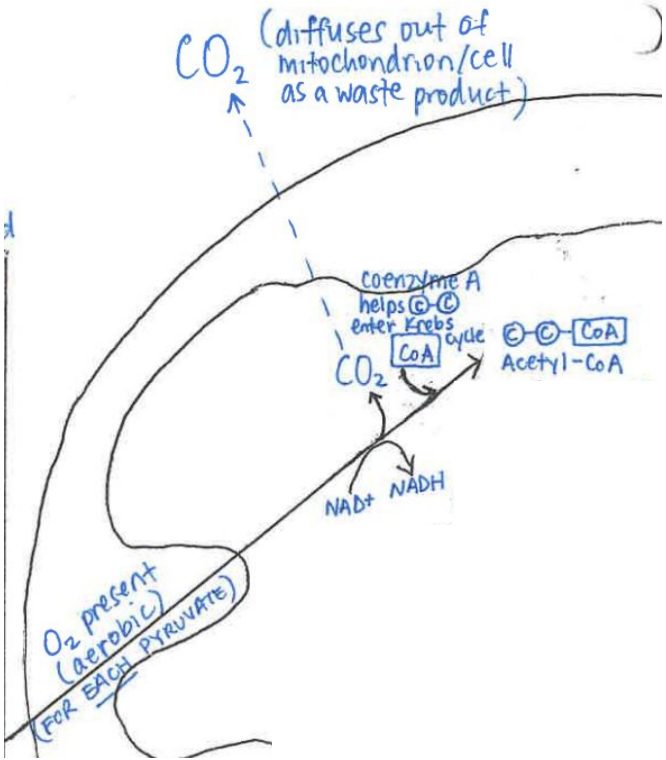
Glycolysis

- takes place in the cytosol (not mitochondria)
- anaerobic!
- glucose -> pyruvate
- products (net): 2 ATP, 2 NADH,
  2 pyruvate/pyruvic acids
- NADH?
  - energy rich coenzymes
  - helps transfer electrons to increase ATP output



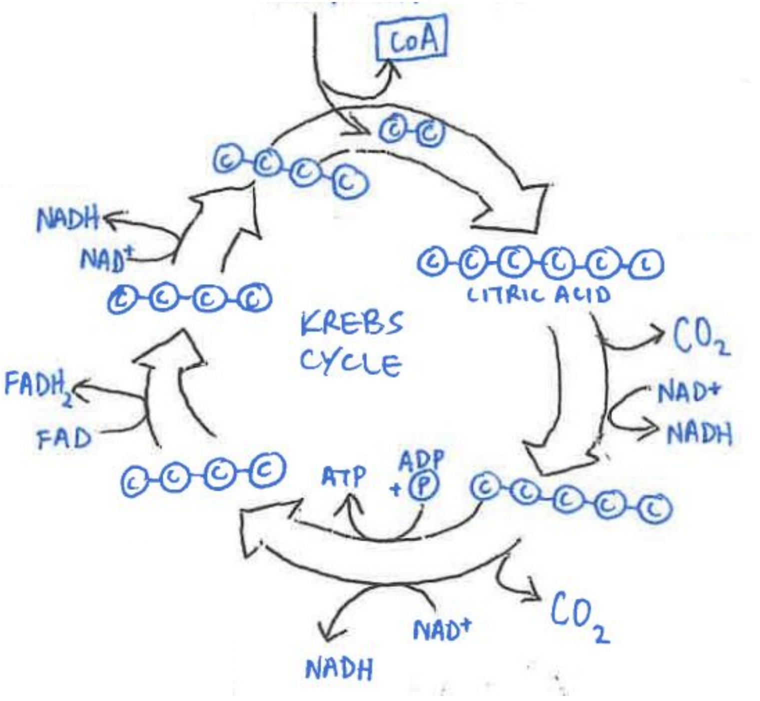
Transport (intermediate step)

- pyruvate **actively** transported into mitochondria (matrix)
- aerobic process
- pyruvate -> acetyl-CoA (2C + CoA)
- CO2 released during this process
- 2 NADH produced (per pyruvate)
- what happened to the third C?
  - became oxidized and was transported out of mitochondria/cell as waste product



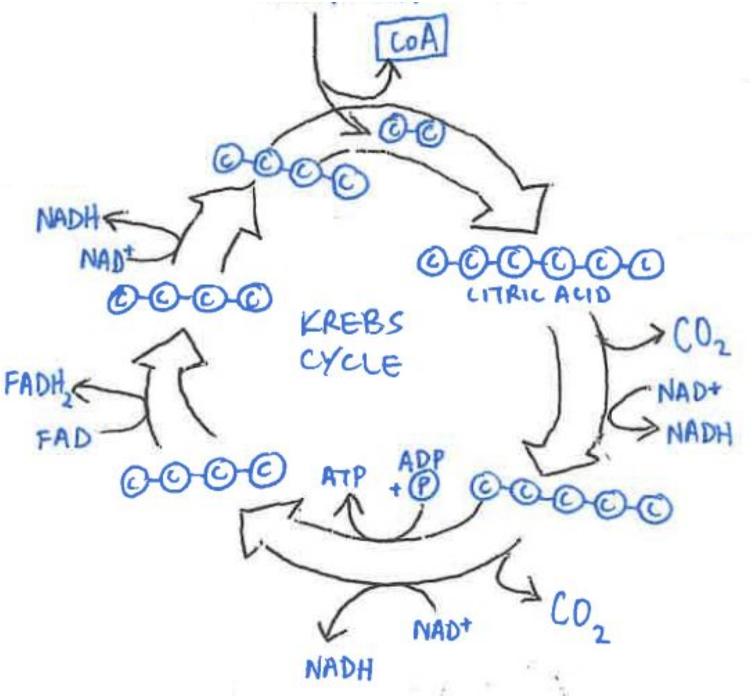
Krebs Cycle

- aka citric acid cycle
- aerobic process
- takes place in the matrix
- CoA transfers the 2C into the Krebs Cycle and leaves
- bonds with the 4C to become 6C (a citric acid)
- 1 C to break off and become CO2
- 1 more C breaks off to become CO2
- now we're left with 4C molecule



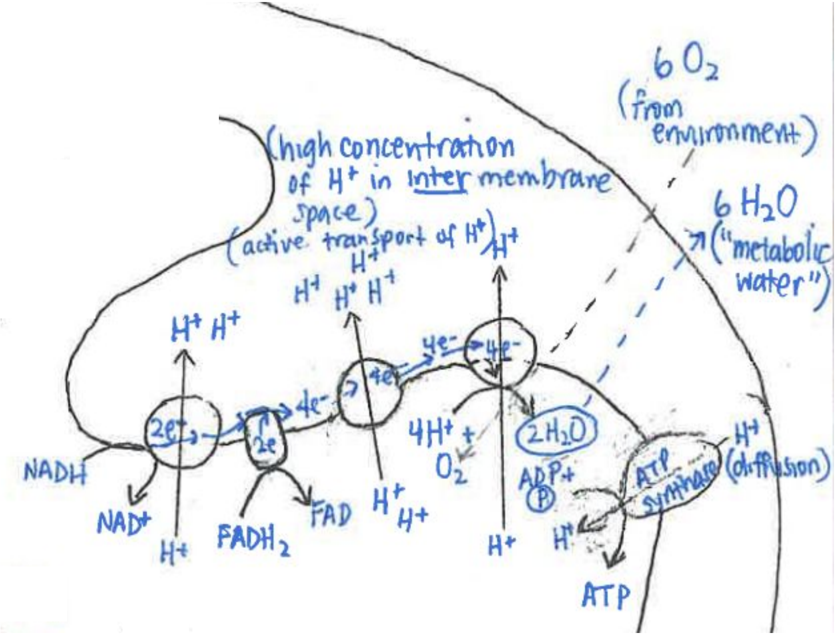
Krebs Cycle (cont.)

- continues through cycle and bonds with the new 2C molecule that enters
- products (PER pyruvate): 3 NADH, 1 FADH2, 1 ATP
- CO2 is byproduct (produced from the cycle but not intentionally)
- goal of Krebs is to produce NADH and FADH2 for ETC



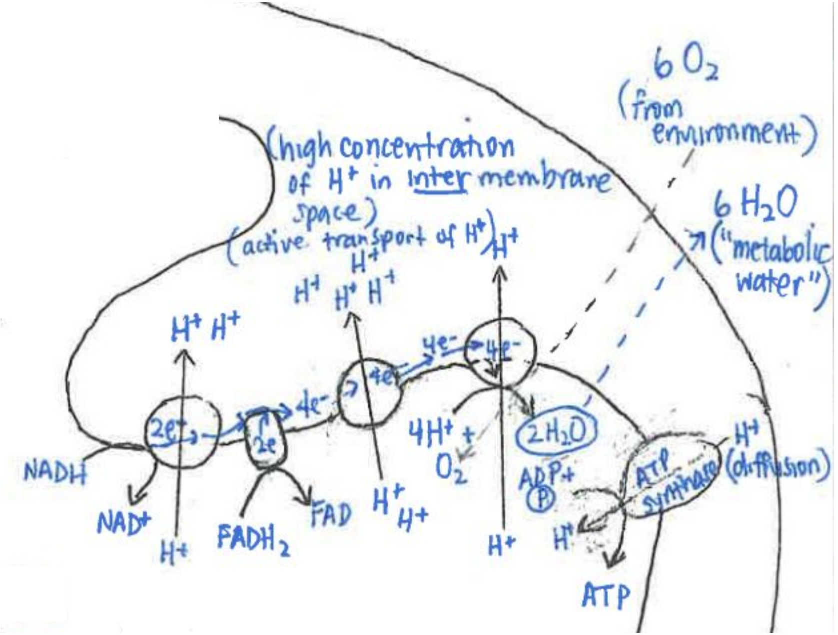
Electron Transport Chain

- along inner membrane of mitochondria
- where most of ATP production actually happens
- when NADH -> NAD+ and FADH2 -> FAD, electrons and H ions are released
- electrons help power the transfer of more H ions outside of the matrix



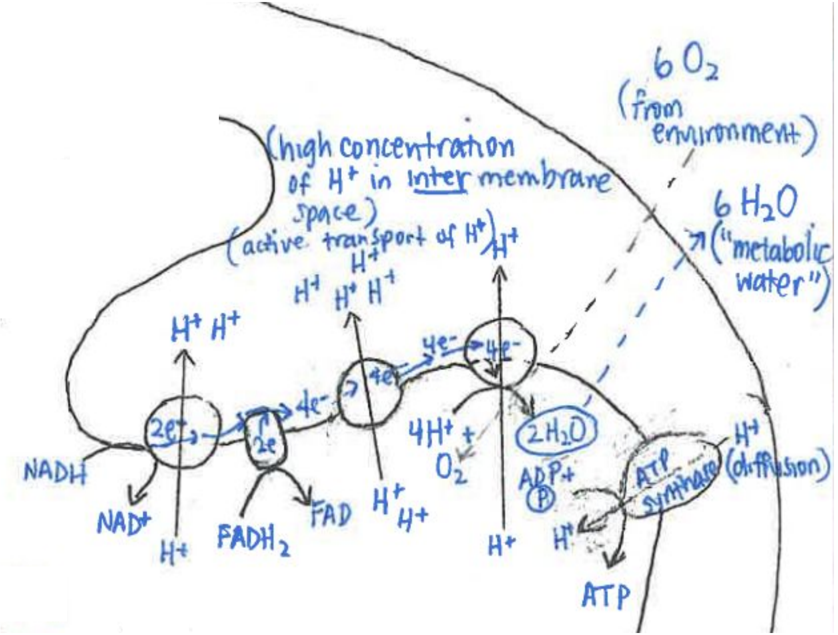
Electron Transport Chain (cont.)

- higher concentration of H+ in intermembrane space causes a concentration gradient
- H+ want to diffuse back into matrix, so they go through ATP synthase (facilitated diffusion)
- ATP synthase gets energy from H+ diffusion to attach ADP and P, making ATP



Electron Transport Chain (cont.)

- final electron acceptor: O2
- when O2 combines with 2H+, you get water (H2O), which diffuses out of the cell
- products of entire ETC: NAD+, FAD, ATP, H2O
- chemiosmosis: pumping of H+ ions down their concentration gradient



Cellular Respiration Overview

$C_6H_{12}O_6 + 6O_2 = 6CO_2 + 6H_2O + ATP$

(glucose) + (oxygen) = (carbon dioxide) + (water) + ATP

<u>net products (can vary!):</u>

glycolysis: 2 ATP, 2 NADH, 2 pyruvate

Krebs (per pyruvate): 3 NADH, 1 $FADH_2$, 1 ATP, 2 CO_2

ETC (per glucose molecule): NAD^+, FAD, 6 H_2O, 26-34 ATP

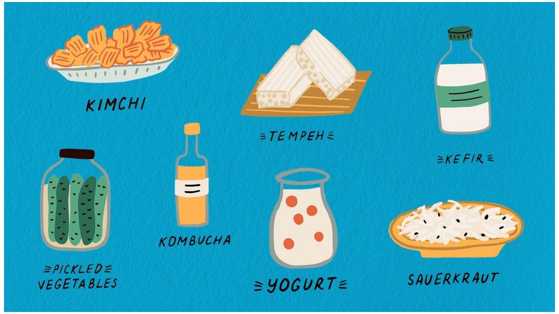
overall: 30-38 ATP

Fermentation

Have you ever felt your stomach hurting after running? Or that you don't have as much energy after a hard workout? Or even how yeast makes barley malt into beer?

Alternative Glucose Breakdown Pathways → Occur when oxygen is not there to act as an acceptor of ETC

- Some animals use other acceptors (ex. sulfate), but we'll focus on fermentation

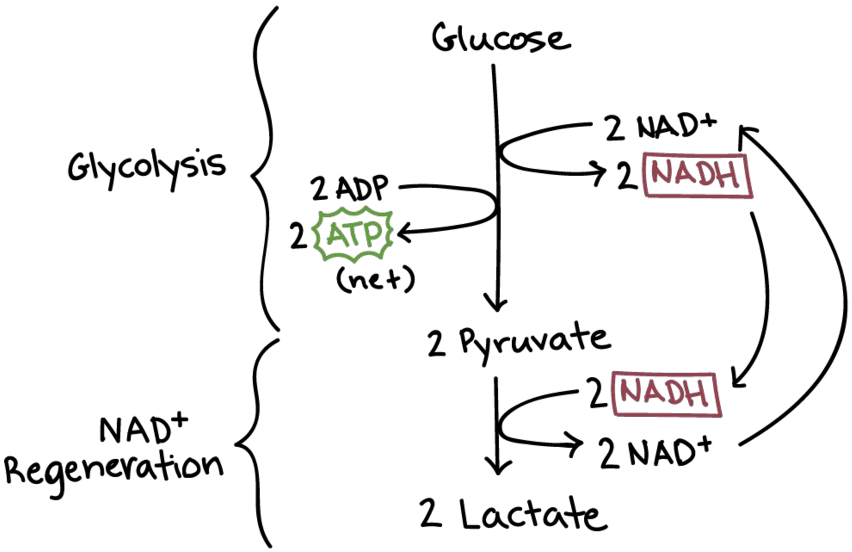


Fermentation vs Cellular Respiration

- Fermentation:
  - Only energy extraction is from glycolysis with a few reactions added at the end
  - Makes a net of 2 ATP compared to CR 30-38 ATP
  - Pyruvate does not continue to citric acid cycle and ETC because etc isn't functional
    - NADH can't drop off electrons and return to NAD+
  - Extra Reactions used to regenerate NAD+ from NADH
    - NADH drops its electrons with another organic molecule
    - Allows steady supply of NAD+

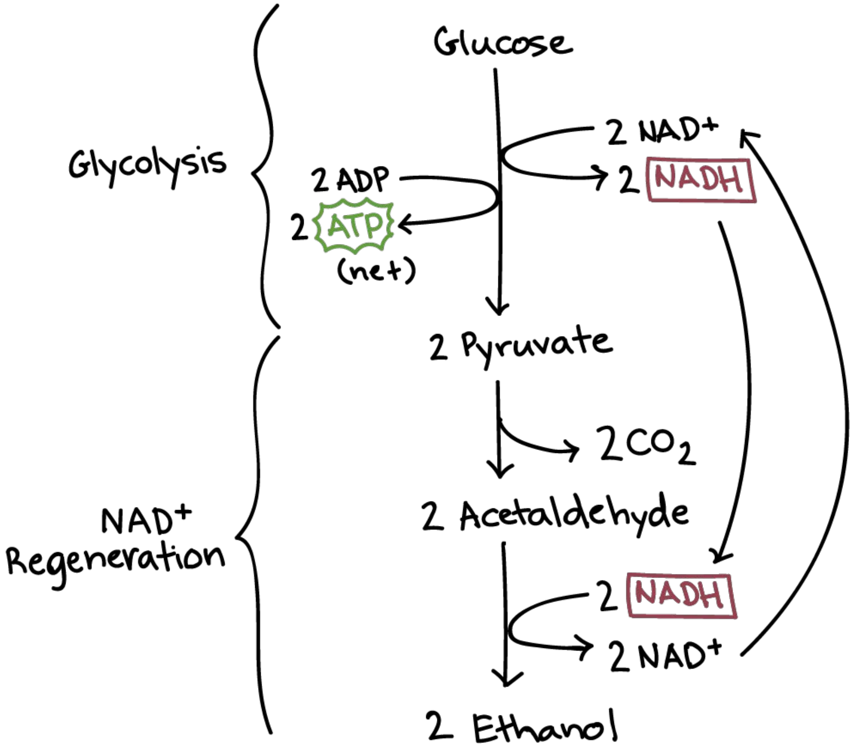
Lactic Acid Fermentation

- NADH transfers electrons directly to pyruvate, making lactate as a byproduct
  - With a hydrogen, lactate becomes lactic acid (yogurt, red blood cells without mitochondria)
- Muscles also do lactic acid fermentation when there is too less oxygen
  - Debate on if lactate causes soreness of exercise
  - Lactate is transported through blood to liver where it is turned back into pyruvate and processed normally
- Done by: Bacteria, Animal Cells
- Results: NAD+, Lactate/Lactic Acid



Alcohol Fermentation

- NADH donates electrons to a different pyruvate molecule → ethanol
- Steps:
  - Carboxyl Group (COOH) is removed from pyruvate and released as CO_2 making a two-carbon molecule called acetaldehyde
  - NADH passes electrons to acetaldehyde making NAD+ again and ethanol as a byproduct
- Results: Ethanol, CO_2, NAD+
- Done By: Bacteria, Plants, Yeast



Fermentation Summary

- Glucose molecules **inefficiently** turned into a net 2 ATP + byproducts
- NAD+ recycled to maintain the cycle
- Done by various organisms when there is no oxygen
  - Bacteria, Animals, Plants
- Glycolysis is the only step that occurs (anaerobic)
- Alcoholic fermentation
  - Glucose → Pyruvate + NADH → Ethanol + CO_2 + NAD+
- Lactic Acid Fermentation
  - Glucose → Pyruvate + NADH → Lactate/Lactic Acid + NAD+

WHY DO PLANTS USE PHOTOSYNTHESIS

BECAUSE THEY WANT A **LIGHT** SNACK : D

Question Bank!

https://docs.google.com/document/d/1RUj5sOnmfIZB2rqpLdSmergL4QoC3jhn9ErsIdSedIA/edit?usp=sharing

THANKS!

ANY QUESTIONS?

Sign in Code